

Clustering Algorithms

BILD62/WI26

March 4, 2026

Aalok Varma, PhD

[pronounced *ah-l-oak*]

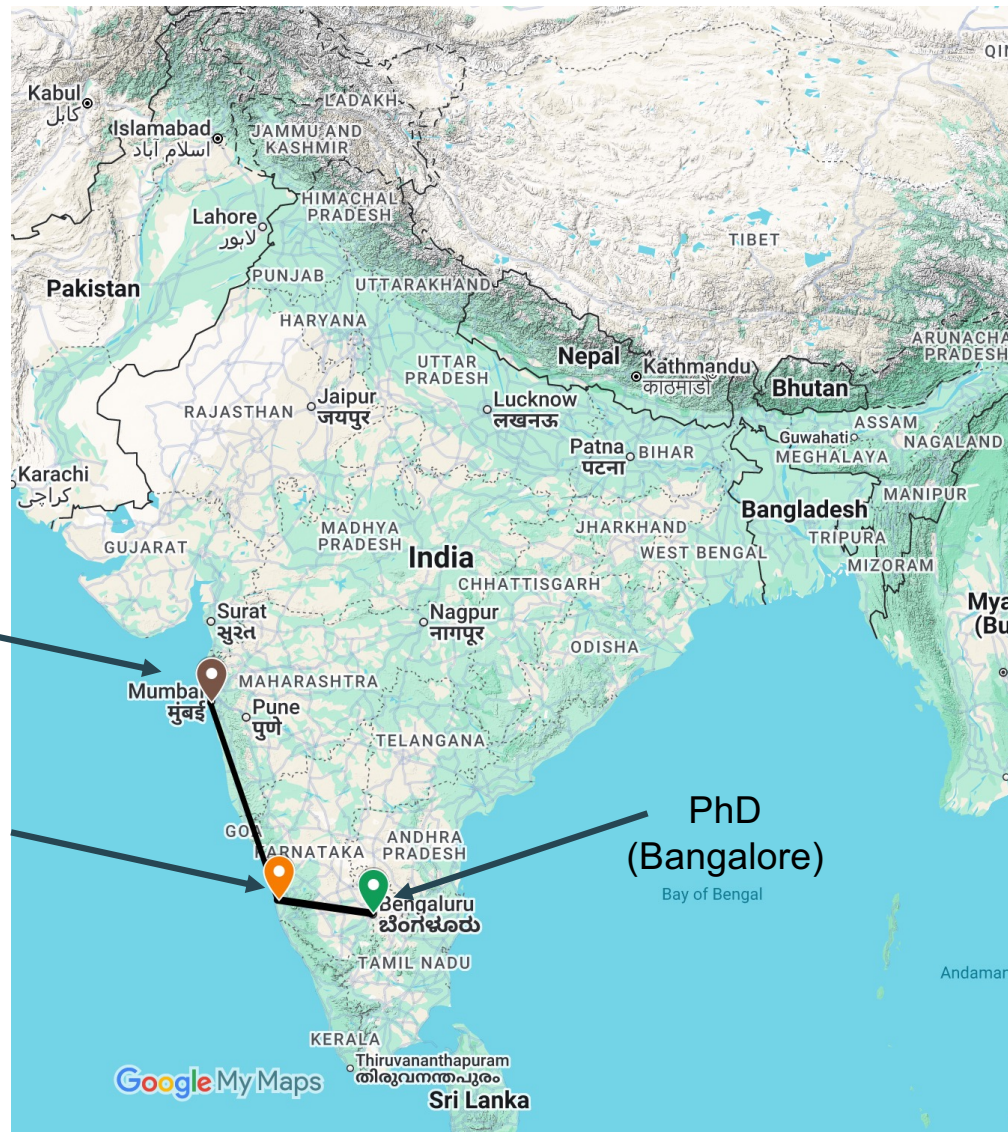
Postdoctoral Fellow, Loren Looger Lab,
Neuroscience Dept, UC San Diego

Who I am

Where I grew up
(Mumbai)

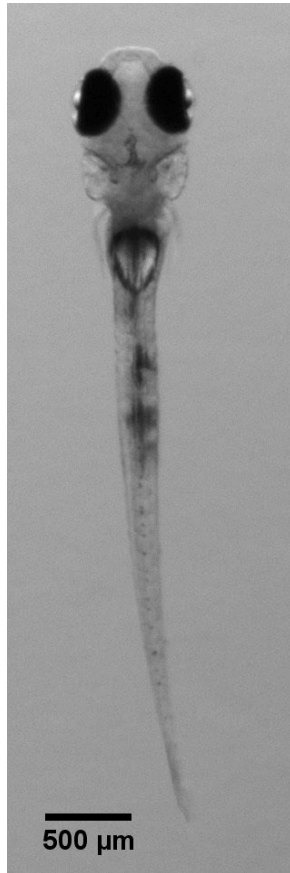
Bachelor's Degree
(Manipal)

PhD
(Bangalore)

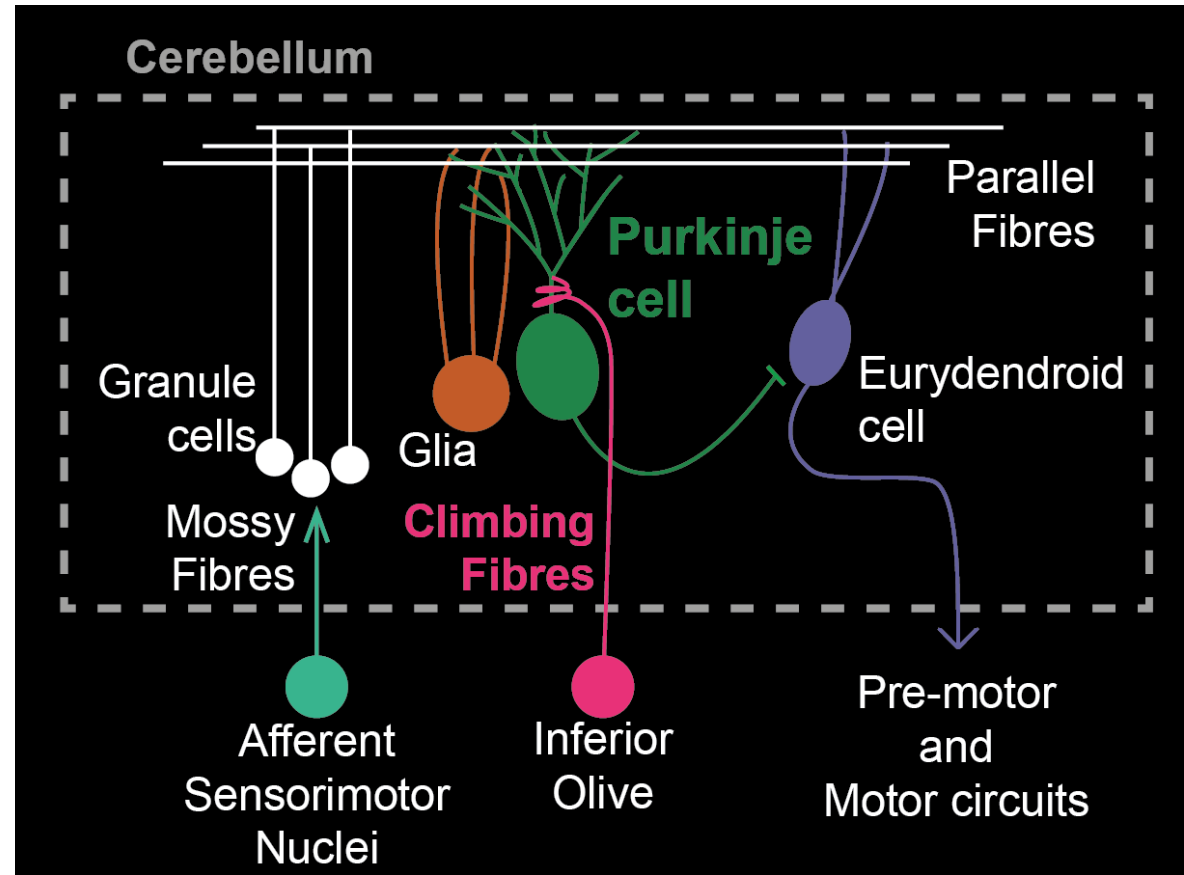


National Centre for Biological Sciences (NCBS)

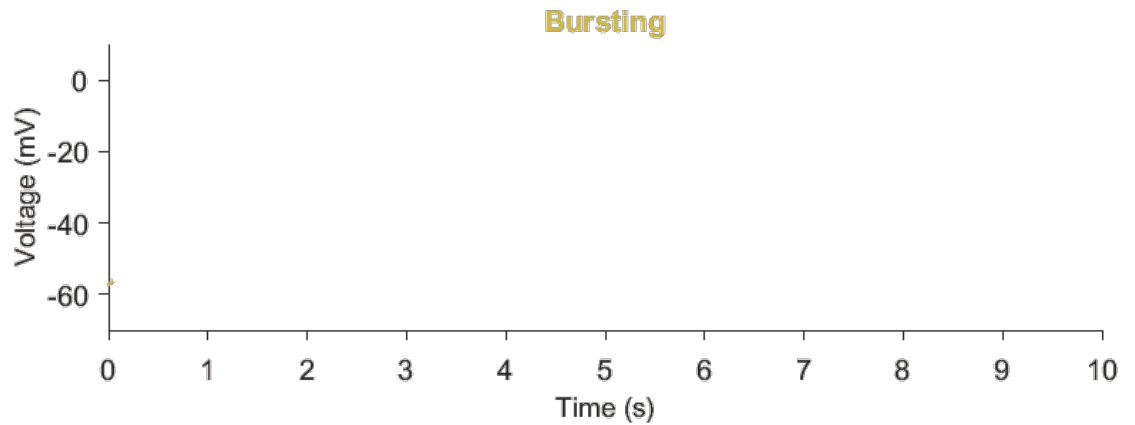
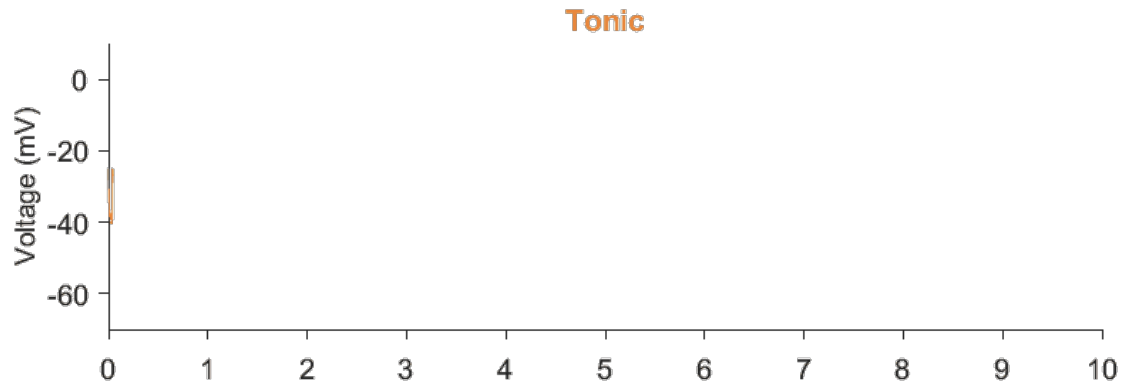


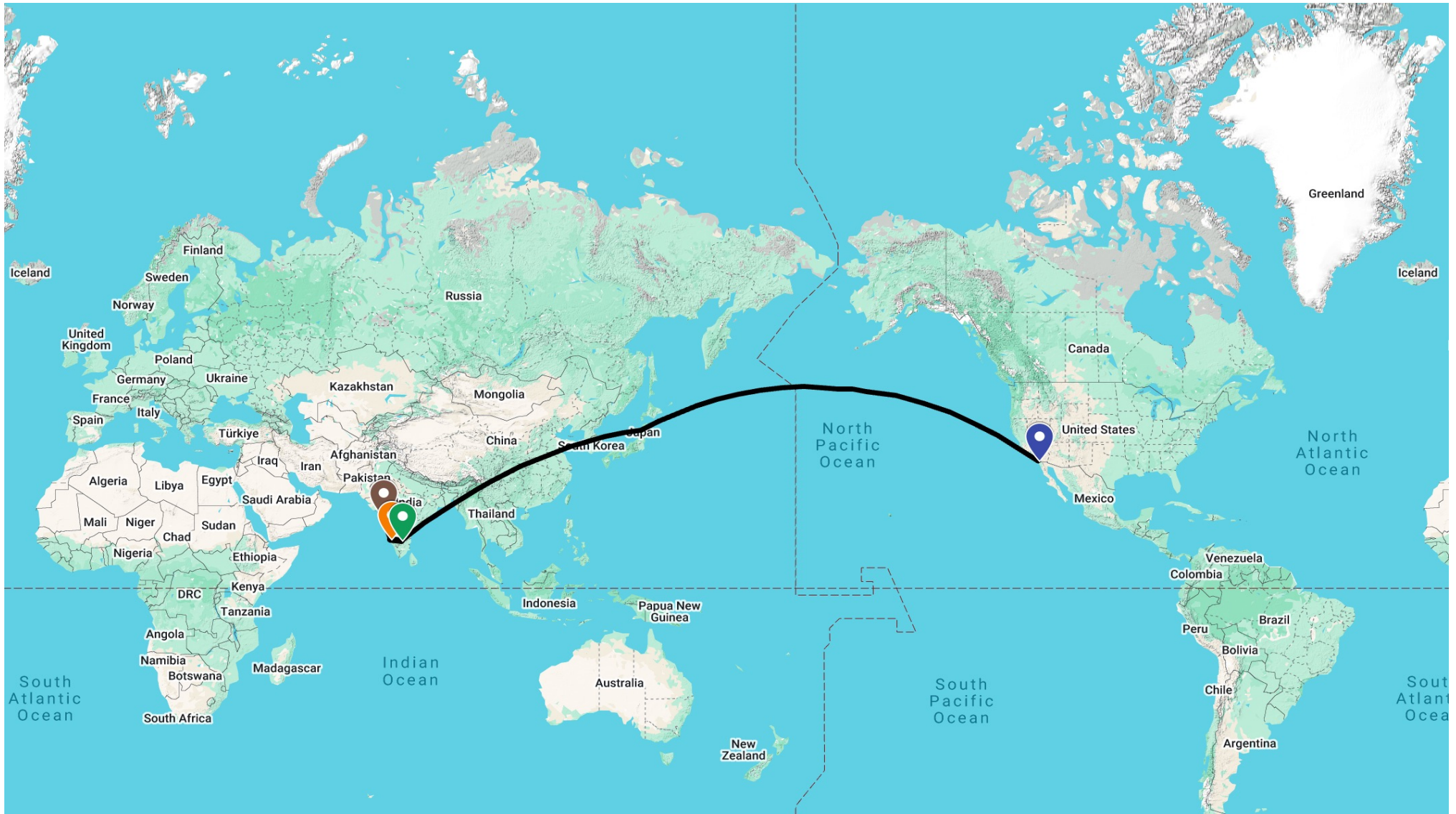


9-day-old
zebrafish (*Danio rerio*)



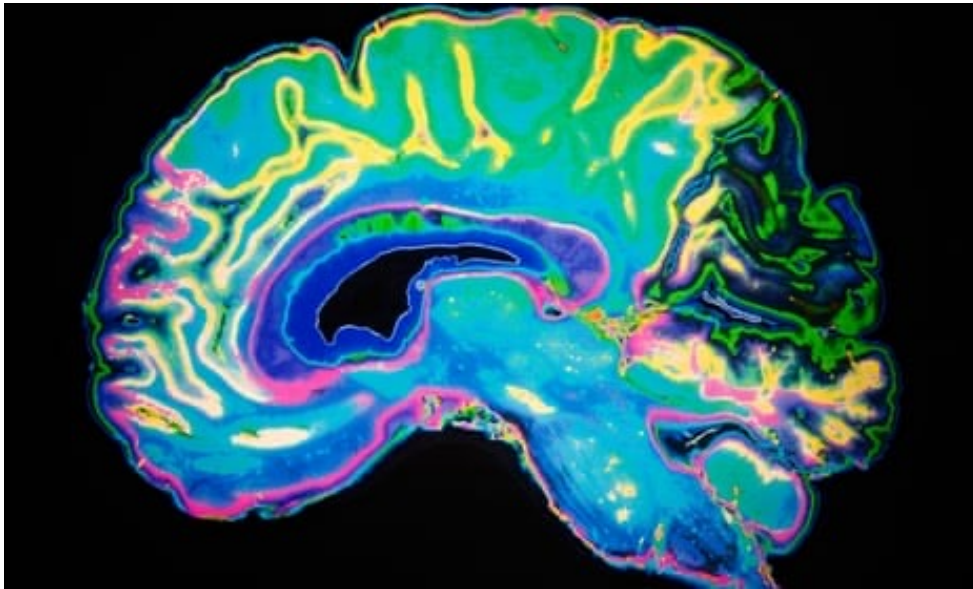
Zebrafish Purkinje neurons are bistable





What I'm working on for my postdoc in the Looger lab

1. Biological effects of climate change



<https://www.theguardian.com/sustainable-business/2014/nov/10/brain-climate-change-science-psychology-environment-elections>

2. Next generation model organisms *Cephalopods*



Euprymna berryi
(Hummingbird bobtail squid)



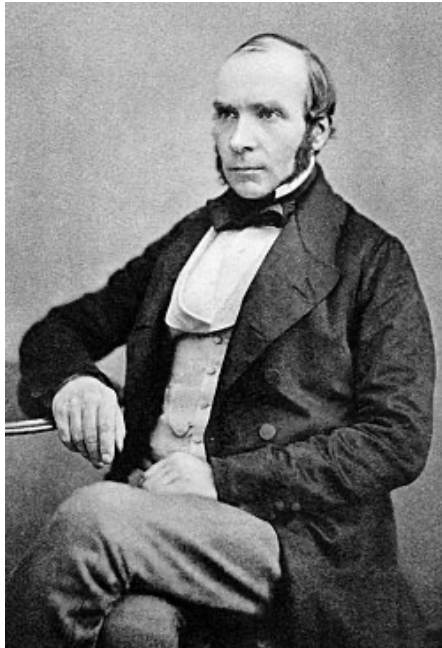
Doryteuthis pealeii
(Longfin inshore squid)

Now for the lecture

Lesson Objectives

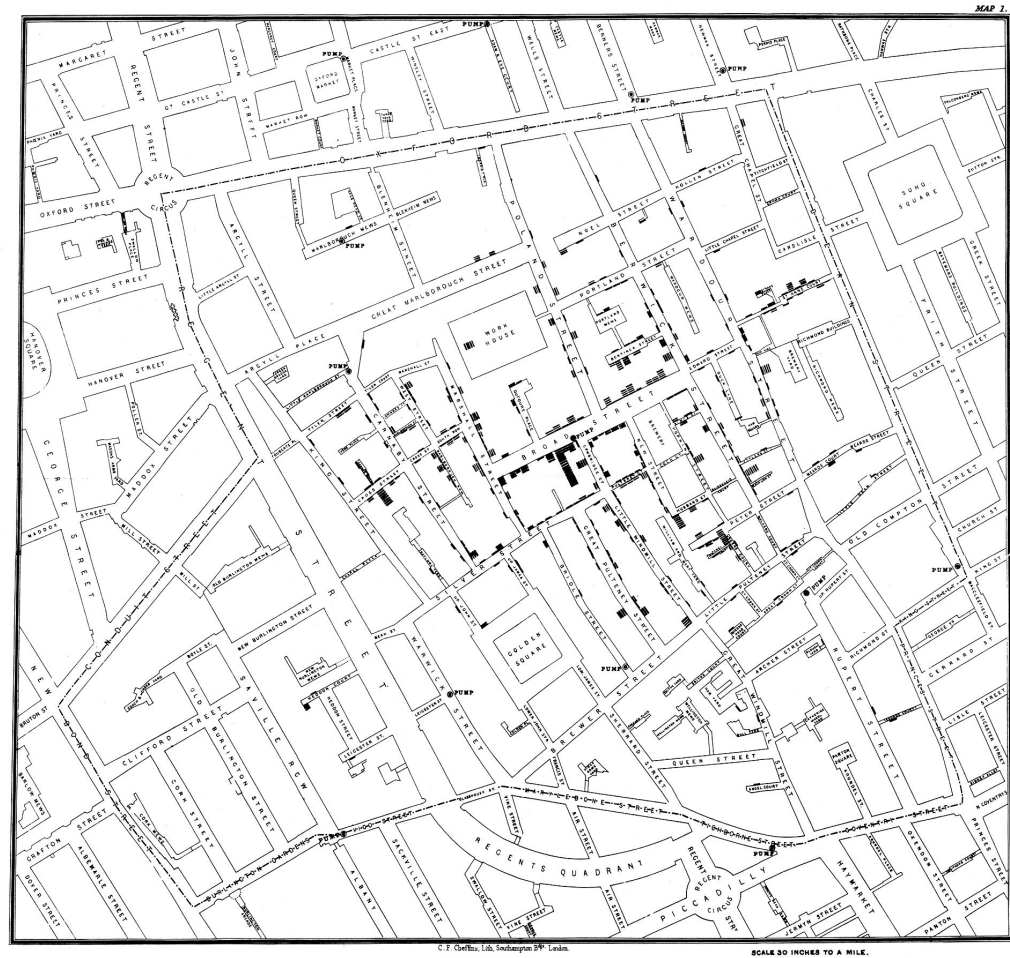
- Recognize a common method of clustering – *k*-means clustering
- Explain how the *k*-means clustering algorithm works
- Compare the speed and complexity of clustering algorithms
- Identify the limitations of the *k*-means clustering algorithm
- Apply *k*-means clustering to data using the *sklearn* package in Python
- Interpret results from clustering analysis

Do you know John Snow?



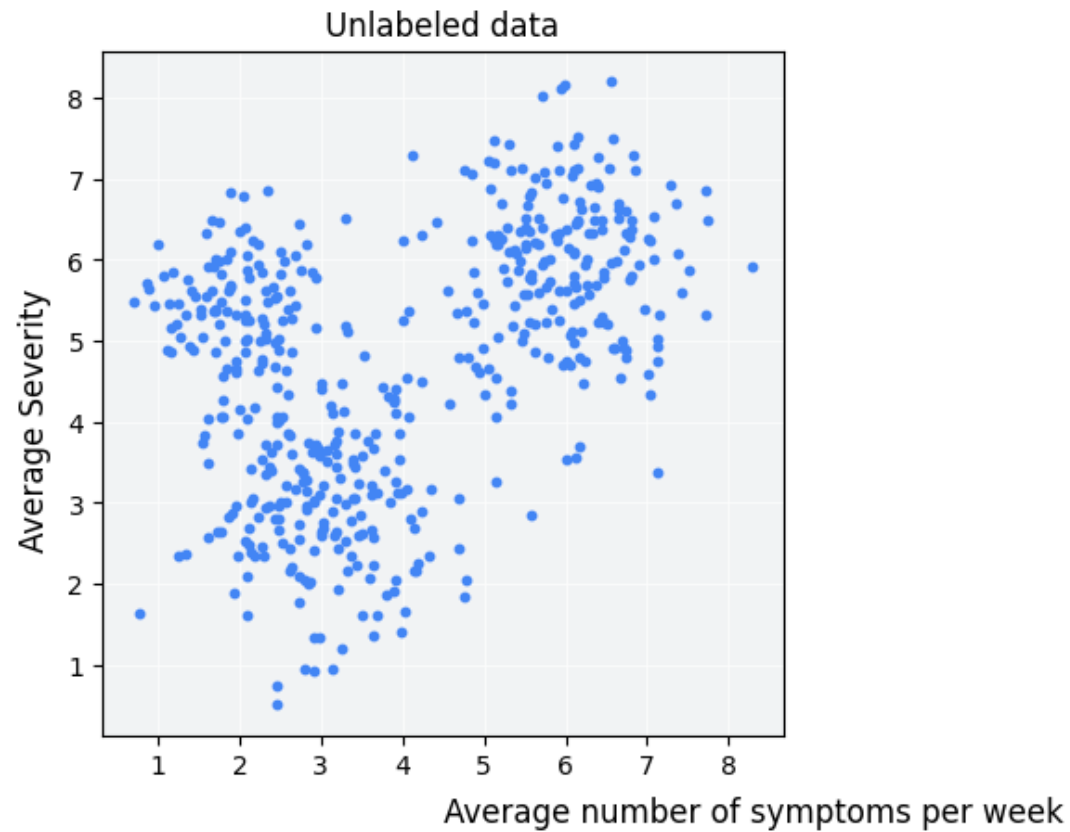
John Snow
1813 – 1858

Image source: Wikipedia

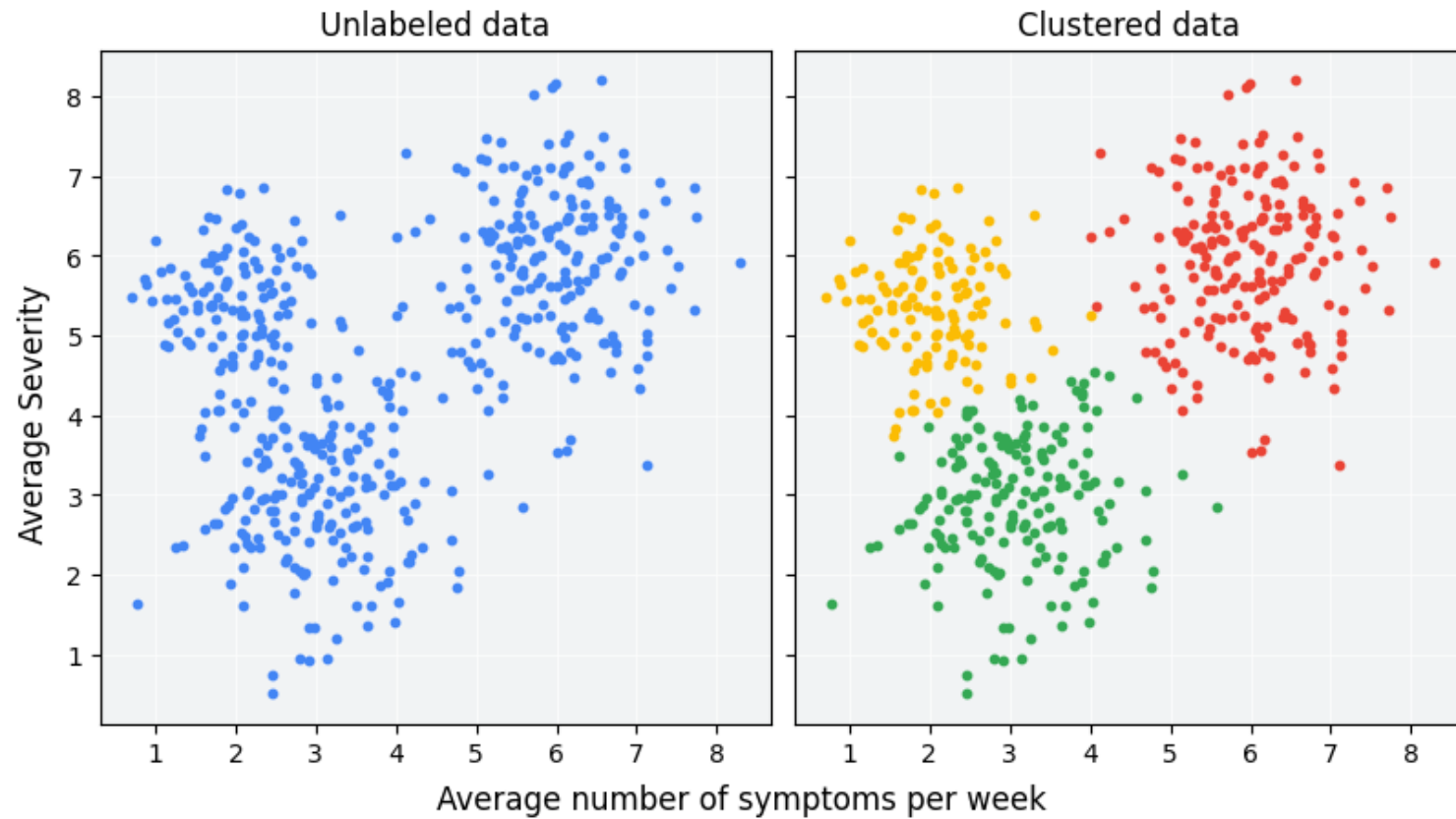


The original map of clusters of cholera cases in London, 1854

Modern clustering looks very different



Modern clustering looks very different



Think-Pair-Share

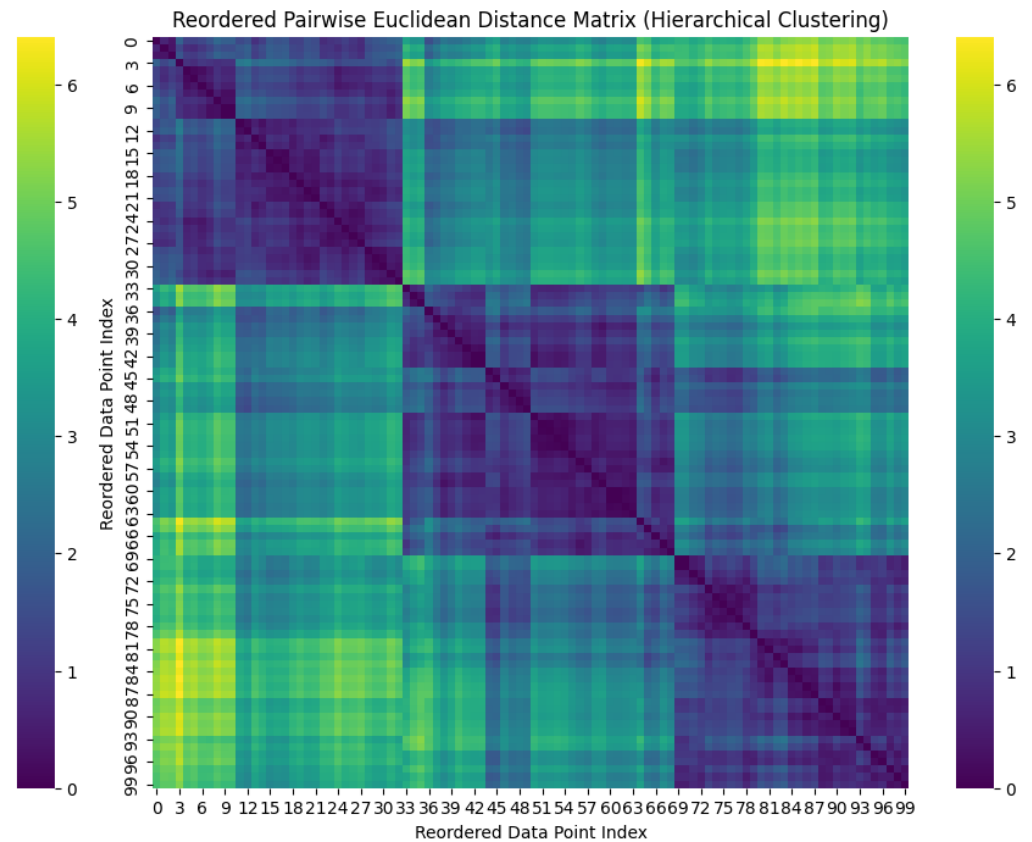
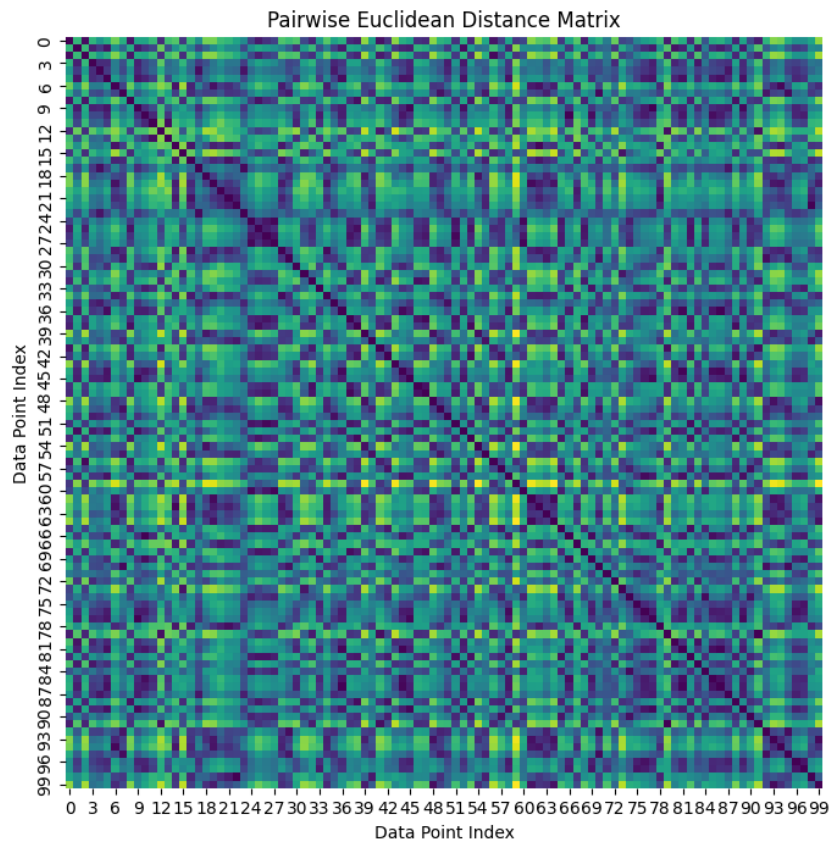
Q: How would you go from unlabelled to labelled data?

What ingredients would you need?

Possible ingredients for clustering algorithm

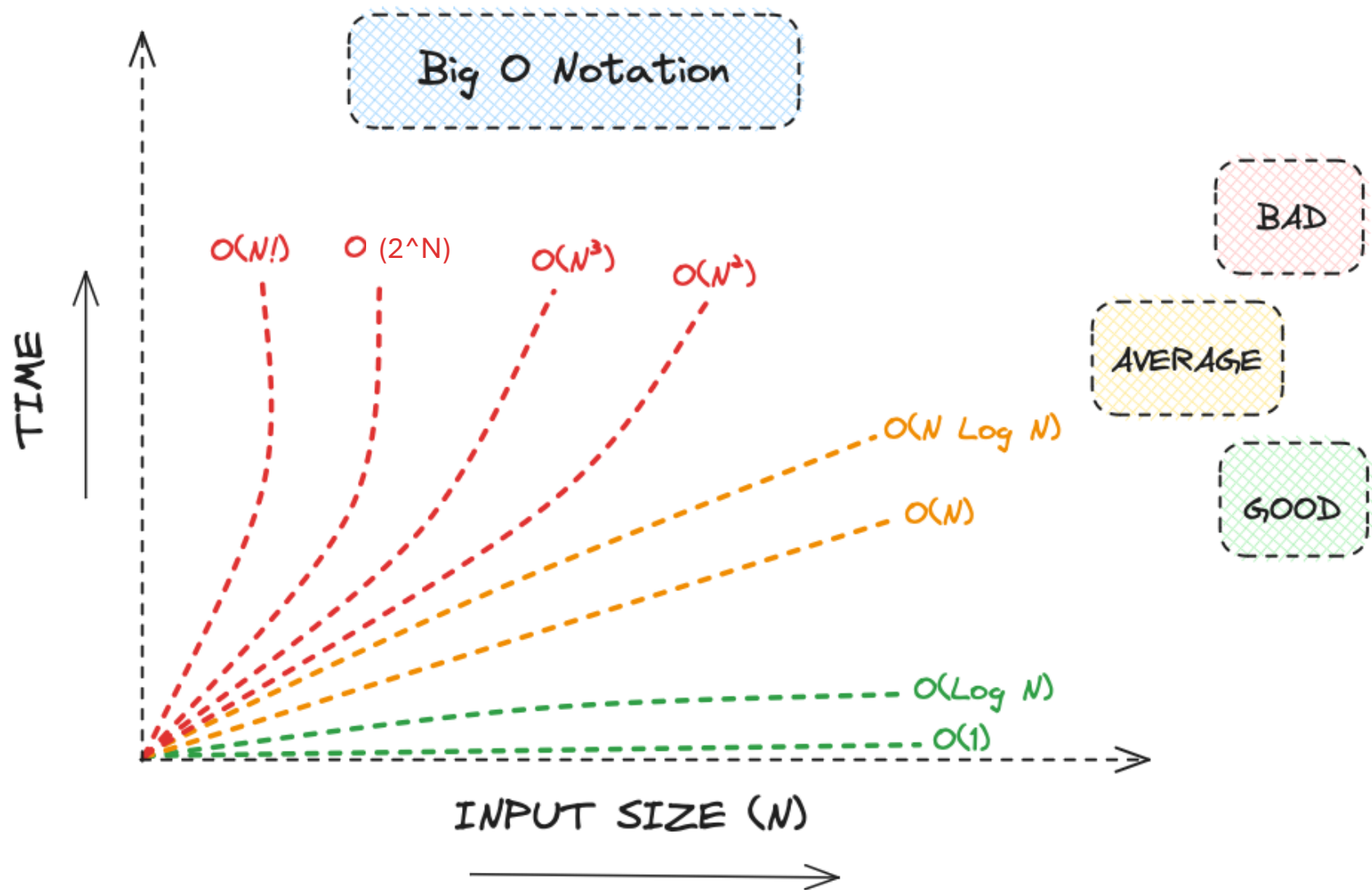
- Some way to define the limits of a cluster
 - Boundary?
 - Centre?
- Some way to group points into clusters
 - Distance between points
 - Distance from points to cluster limits
 - Thresholds?

A naïve idea for clustering



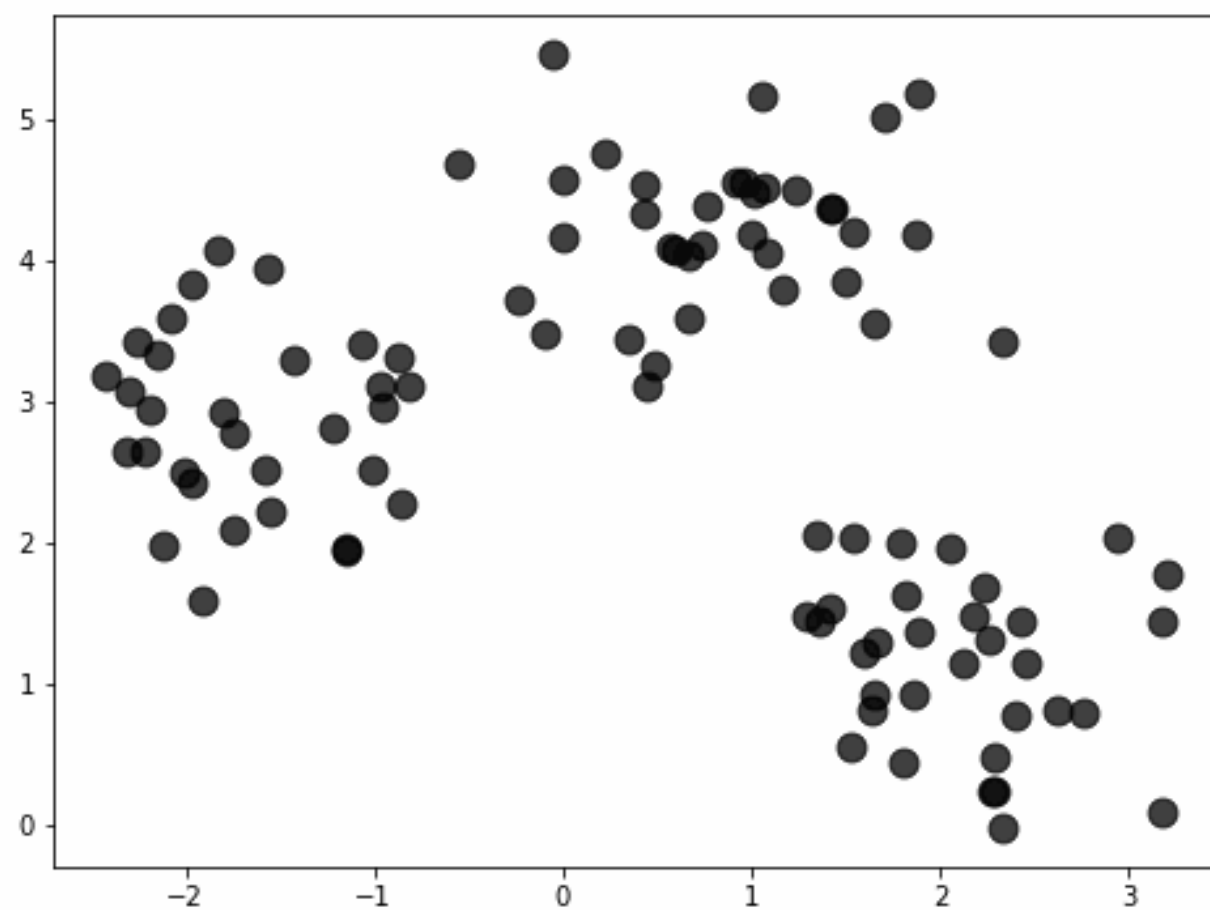
How slow/fast is this algorithm?

- You have to look at worst-case scenarios
- Big-O notation for computational complexity



Let's do better – k -means clustering

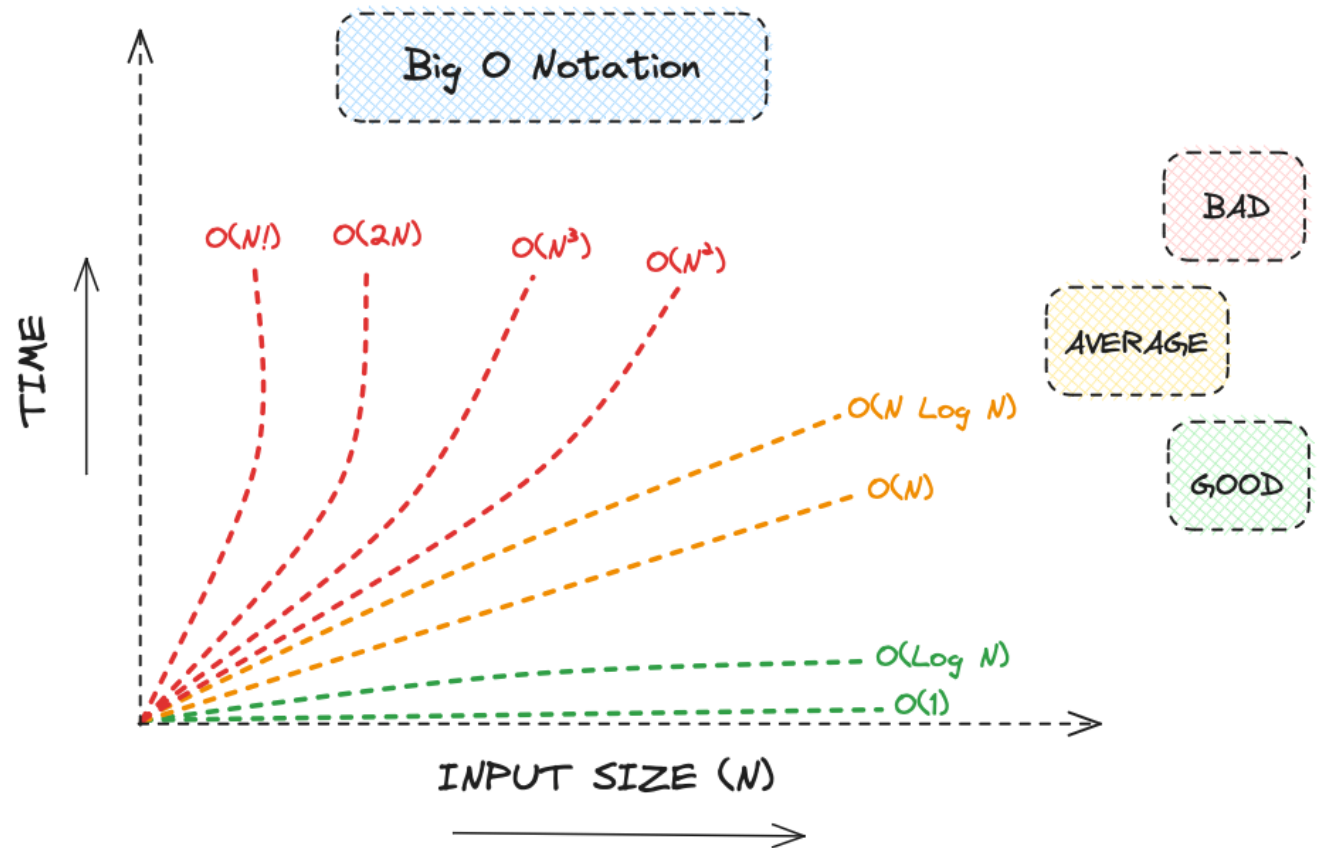
- Pre-algorithm
 - Define each cluster by a cluster center – *centroid*
 - Decide a value of k (i.e. how many clusters are in the data?)
- Initialization
 - Choose k random points as cluster centers
- Algorithm:
 - 1 - Calculate distance from data to each cluster center
 - 2 - Assign data to cluster based on proximity
 - 3 - Define new cluster centers as the ***mean*** of data assigned to the cluster
 - Repeat steps 1-3 until cluster centers don't change anymore (i.e. the algorithm "converges")



How fast is *this* algorithm?

$O(n*k*d)$

Effectively $O(n)$



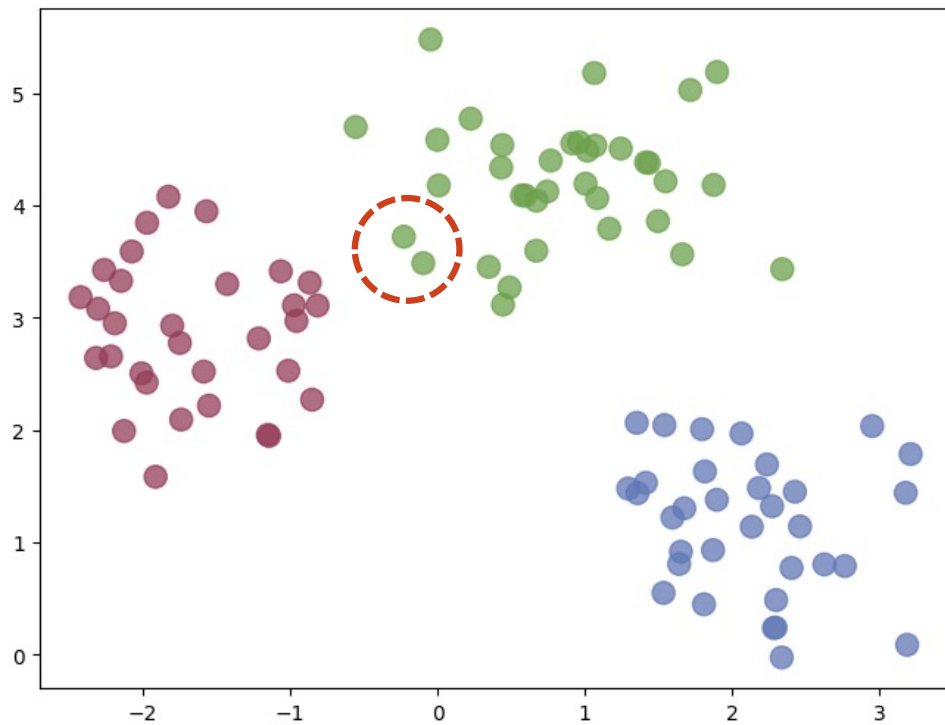
Hands-on coding time

Notebook 1

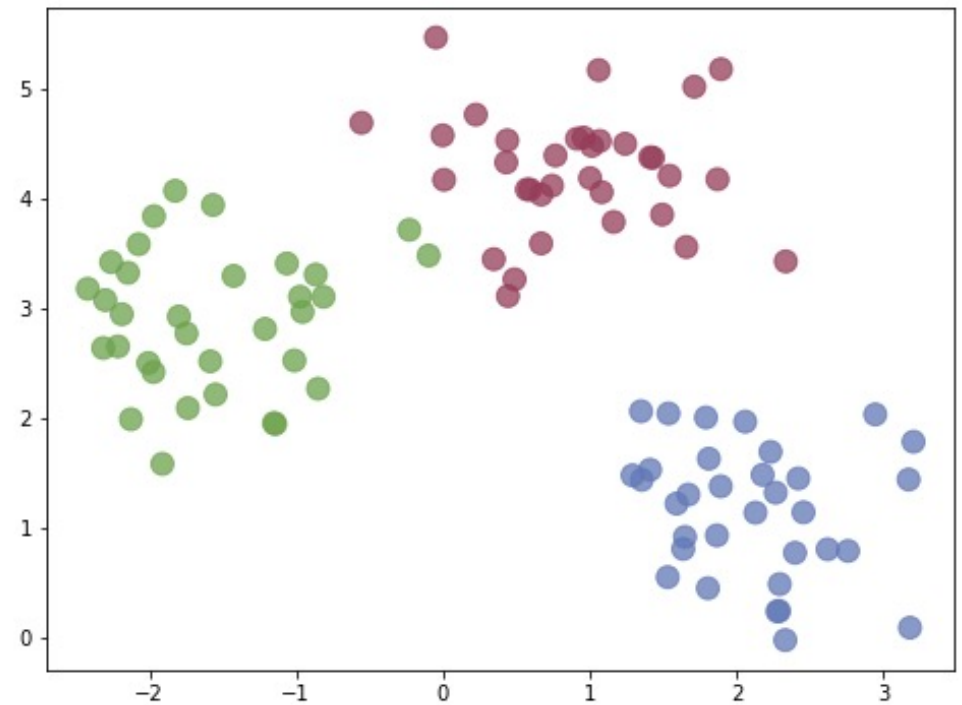
Break time

k -means clustering isn't perfect

Clusters from k -means

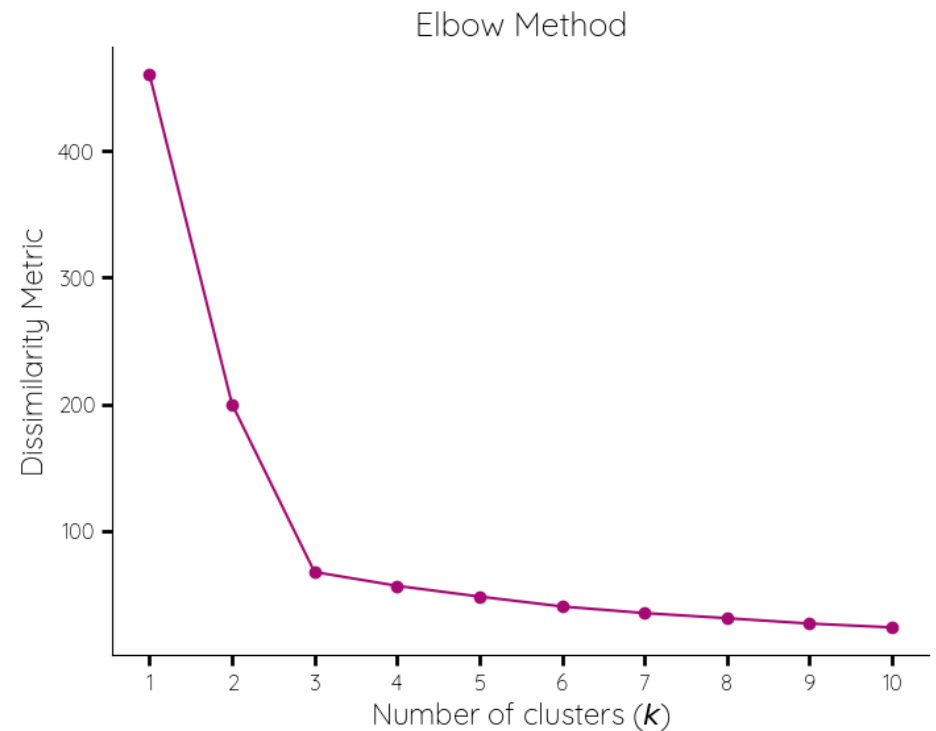


True grouping (i.e. "**Ground Truth**")



How do you choose k ?

1. Advance knowledge
2. “Elbow” method
 - *Hands-on coding time*
3. Gap statistic
 - *Slightly advanced – beyond scope of this lecture*



Other modifications/improvements

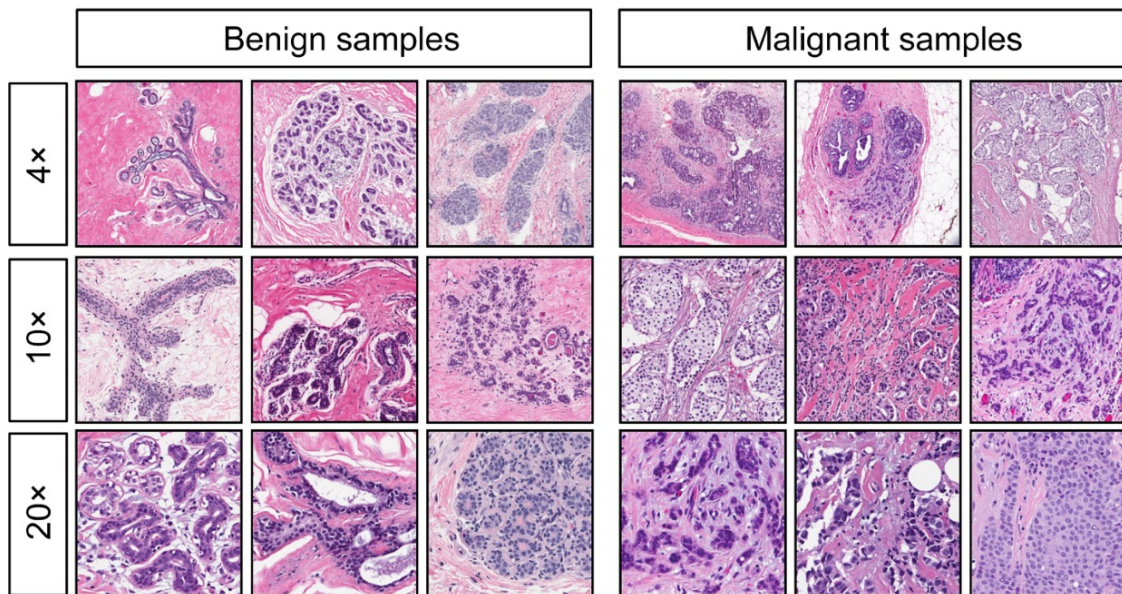
- Figure out better starting points
 - *k-means++* [This is the default method in *sklearn*]
- Replace “mean” with other measures of central tendency
 - *k-medians*
- “*Fuzzy*” clustering
 - Datapoints could belong to more than one cluster

Let's work with some real data

Notebook 2

Step 0 – Know the data!!

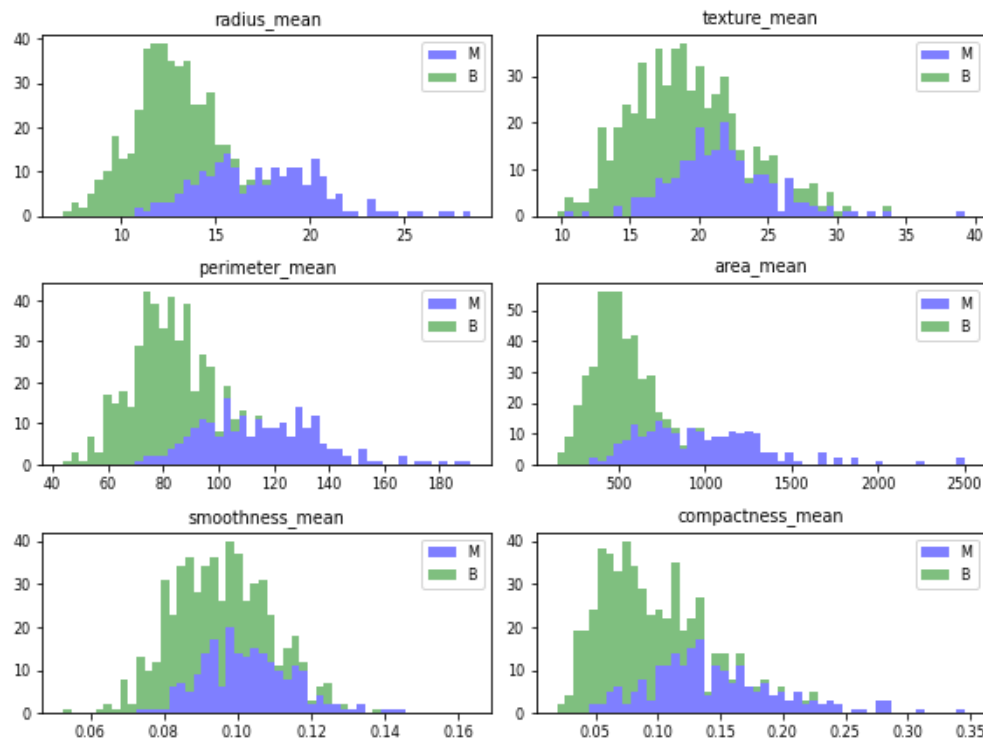
- Source: Wisconsin Breast Cancer Data
- Fine needle aspirate (FNA) of a breast mass



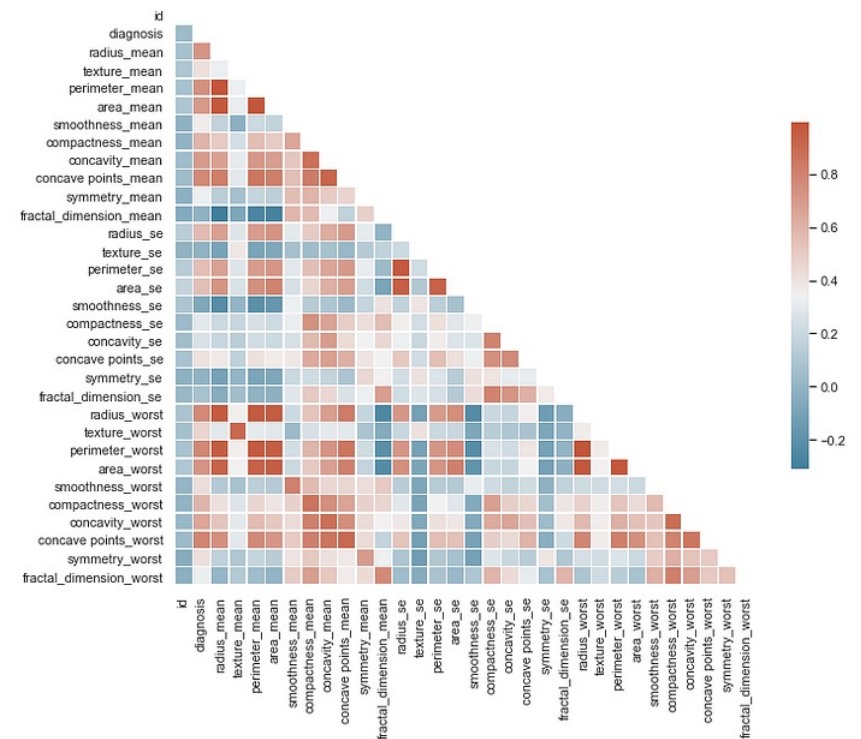
Extracted image parameters:

- Radius
- Perimeter
- Area
- Smoothness
- Texture
- ...

Step 1 – Explore the data



Histograms to look at distributions



Correlations between variables

Step 2 – Preprocess the data (if needed)

- For today, I have run "Principal Components Analysis" (PCA) on a subset of the data

Step 3 – Cluster

- Hands-on coding time...

Some example real-world applications of clustering

1. Neuroscience

- Hierarchical clustering of neural activity imaging data

2. RNAseq analysis

- Find groups of genes that are co-regulated

3. Image/video segmentation

- k -Nearest Neighbour (kNN) algorithm to group pixels into objects

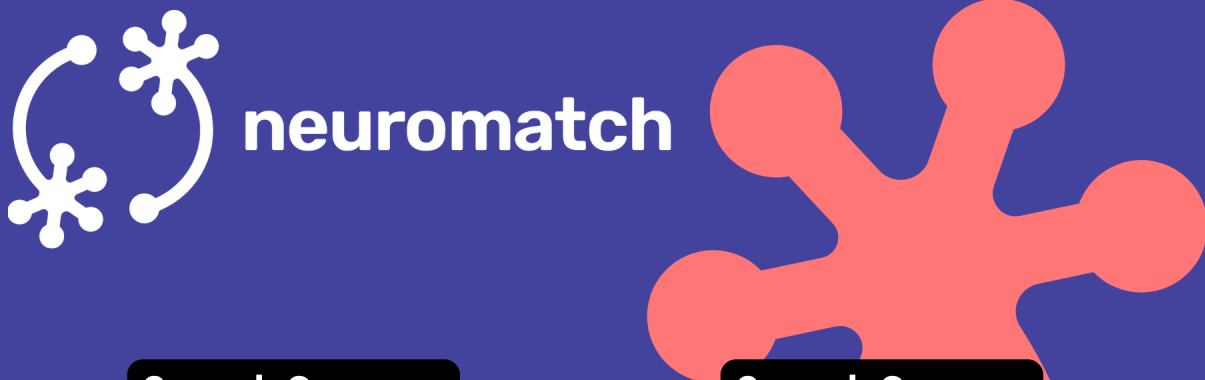
Summary of what we learned today

1. Clustering is useful to group and/or classify data
2. Elements of a clustering algorithm
 - a. Cluster definition
 - b. Distance measure
3. k-means clustering is a fast, efficient way to cluster data
 - a. Takes a few milliseconds (timed using *timeit*)
 - b. “Linear”-ish using big-O notation: $O(n*k*d)$
4. k-means is guaranteed to converge, but ***not*** to a *unique* solution
5. Choosing *k* is somewhat subjective
 - a. Use prior domain-specific knowledge
 - b. “Elbow” method

Some ideas for further exploration

- Fix the labels of K-means to make sure they're correct
- Try out some different toy datasets
 - *sklearn* has some
- Explore some real datasets
 - Kaggle – Website with data science challenges
 - Ask me/other lab folks for some real data
- Try other clustering algorithms

Neuromatch Academy courses

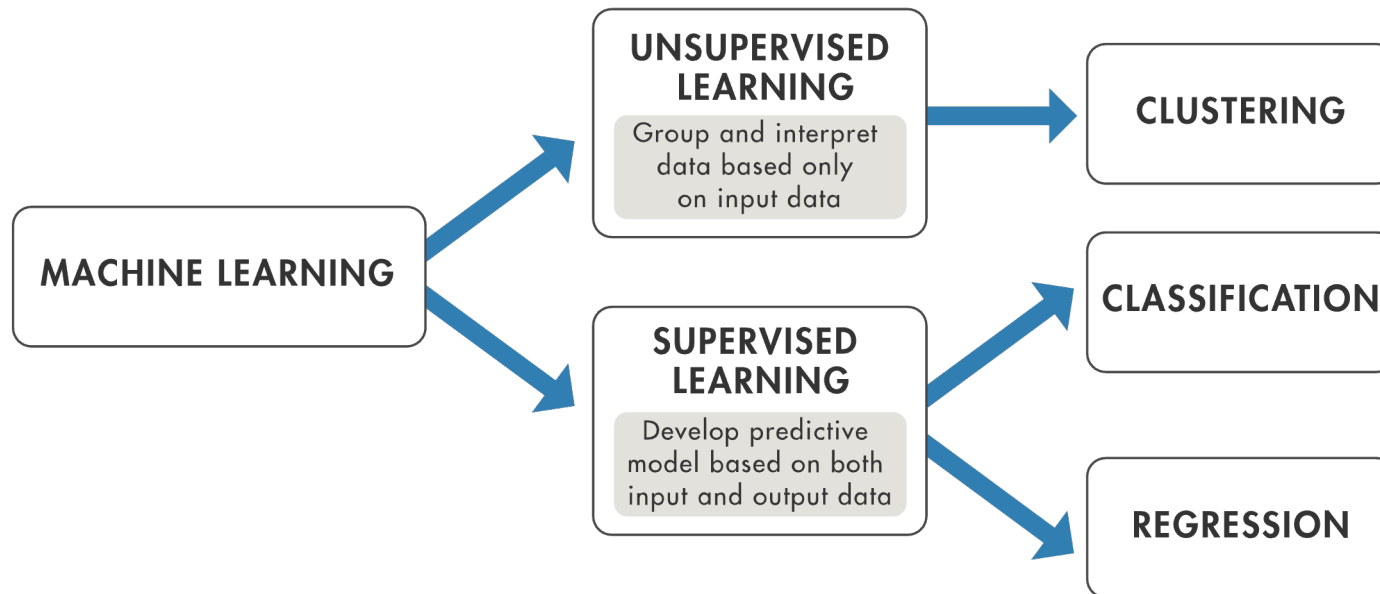


The graphic features the Neuromatch logo on the left, which consists of two stylized neuron icons connected by curved lines. To the right of the logo is the word "neuromatch" in a white, lowercase, sans-serif font. Further right is a large, red, stylized neuron icon with multiple branching processes. Below these elements, the course schedule is presented in a grid-like format with two main categories: "3-week Courses" and "2-week Courses". Each category has a black header with white text. Under the "3-week Courses" header, there are two white boxes. The first box contains the text "Computational Neuroscience" and "6-24 July". The second box contains the text "Deep Learning" and "6-24 July". Under the "2-week Courses" header, there are two white boxes. The first box contains the text "NeuroAI" and "13-24 July". The second box contains the text "Comp Tools for Climate Science" and "13-24 July". The entire graphic is set against a dark blue background.

3-week Courses		2-week Courses	
Computational Neuroscience	Deep Learning	NeuroAI	Comp Tools for Climate Science
6-24 July	6-24 July	13-24 July	13-24 July

Apply by March 15, 2026

You just did some **machine learning**!!



Extra resources

1. Introduction to Computation and Programming Using Python – John Guttag
2. <https://ocw.mit.edu/courses/6-0002-introduction-to-computational-thinking-and-data-science-fall-2016/resources/lecture-12-clustering/>
3. <https://www.mathworks.com/discovery/clustering.html>
4. <https://www.huber.embl.de/msmb/05-chap.html#what-are-the-data-and-why-do-we-cluster-the>
5. <https://scikit-learn.org/stable/modules/clustering.html#k-means>
6. <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>
7. https://scikit-learn.org/stable/auto_examples/cluster/plot_cluster_comparison.html
8. https://seaborn.pydata.org/tutorial/color_palettes.html